Question	Answer	Marks	Partial Marks
1(a)(i)	Translation $\begin{pmatrix}-7\\-1\end{pmatrix}$ oe	**2**	**B1** for each
1(a)(ii)	Rotation 90° clockwise oe (5, 1)	**3**	**B1** for each
1(b)(i)	Image at (2, 6) (3, 6) (3, 8)	**2**	**B1** for reflection in $y = k$, $k \neq 2$ or for reflection in $x = 2$
1(b)(ii)	Image at (–4, 4) (–6, 4) (–6, 8)	**2**	**B1** for an enlargement, sf –2 in the wrong position
2(a)	1960	**2**	**M1** for $\frac{1}{2} \times 9.8 \times 20^2$ oe
2(b)	1.5 or 1½ or $\frac{3}{2}$	**3**	**M1** for a first correct step, e.g. $20y - 15 = 15$ or $4y - 3 = 3$ **M1FTdep** for a second correct step, e.g. $20y = 30$ or $4y = 6$ or $y - \frac{15}{20} = \frac{15}{20}$ oe
2(c)	$9x - 10$ final answer	**2**	**B1** for $kx - 10$ or $9x + c$ or **M1** for $15x - 24$ or $-6x + 14$ or **B1** for correct answer seen and then spoiled
2(d)	$\sqrt[3]{\frac{2b^2 - A}{3}}$ oe final answer	**3**	**M1** for isolating $3c^3$, $3c^3 = 2b^2 - A$ oe or for $\frac{A}{3} = \frac{2b^2}{3} - c^3$ or $\frac{A}{-3} = \frac{2b^2}{-3} + c^3$ **M1FT** for isolating c^3, follow through their first step dep on a 3-term expression with a kc^3 term **M1FT** taking the cube root to the final answer, follow through their previous step Maximum of two marks if answer incorrect
2(e)	$(2q - 1)(3p - 2)$ or $(1 - 2q)(2 - 3p)$ final answer	**2**	**M1** for $2q(3p - 2) - [1](3p - 2)$ or $3p(2q - 1) - 2(2q - 1)$ or for correct answer seen then spoiled
3(a)(i)	227 900 000	**1**	

Question	Answer	Marks	Partial Marks
3(a)(ii)	51 200 or 51 190 or 51 194	**2**	**M1** for $\frac{35.8}{100} \times 143\,000$ After 0 scored SC1 for answer figs 512 or figs 5119 or figs 51194
3(a)(iii)	2.43 or 2.434…	**1**	
3(a)(iv)	3000 or 3004 to 3005	**2**	**M1** for $\frac{4.495\times10^9}{1.496\times10^8}$ [× 100] oe After 0 scored SC1 for answer figs 3 or figs 3004…. or figs 3005
3(a)(v)	1.52 or 1.522…	**2**	**B1** for 1AU = 1.5[0] × 10^8 or 1.497… × 10^8 [km] or $1\text{km} = 6.68\times10^{-9}$ *or* $6.678\ldots\times10^{-9}$ [AU] OR **M1** for $\frac{5.2\times2.279[\times10^8]}{7.786[\times10^8]}$ oe After 0 scored **SC1** for answer figs 152 or figs 1522……
3(a)(vi)	4890 or 4885…	**2**	**M1** for $d \times \left(1+\frac{39.2}{100}\right)$ = 6800 oe
3(b)(i)	2.9979 × 10^5 × 60^2 × 24 × 365.25	**M1**	After **M0** **SC1** for 2.9979 × 10^5 × 31557600 oe
	= 9.4606… × 10^{12}	**A1**	
3(b)(ii)	2.54 or 2.536 to 2.537	**2**	**M1** for $\frac{2.4\times10^{19}}{9.461\times10^{12}}$ oe
4(a)(i)	$\frac{2}{5}$ $\frac{5}{9}$, $\frac{4}{9}$ $\frac{5}{9}$, $\frac{4}{9}$	**2**	**B1** for $\frac{2}{5}$ and a pair of probabilities for spinner B that sum to 1
4a(ii)(a)	$\frac{1}{3}$ oe	**2**	**FT dep** *their* tree diagram **M1** for $\frac{3}{5}$ × *their* $\frac{5}{9}$
4a(ii)(b)	$\frac{2}{3}$ oe	**1**	**FT** dep 1 – *their* $\frac{1}{3}$
4(b)	72	**1**	